

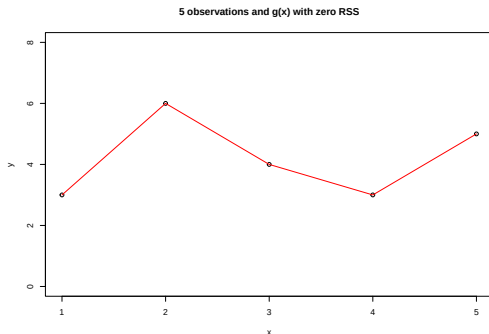
## Smoothing splines and Generalized Additive Models (GAM)

# Smoothing splines

$$y_i = f(x_i) + \varepsilon_i, \quad f \text{ any function}$$

Find the  $\hat{f}$  that minimizes

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$



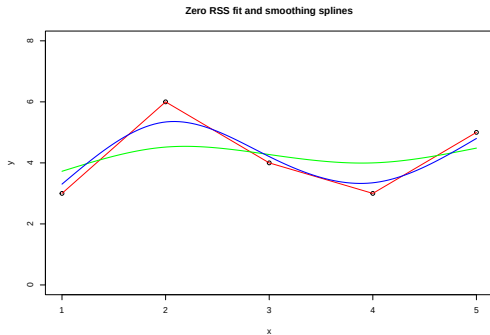
## Smoothing splines

- ▶ The function above is continuous but non-differentiable. We want the function to be smooth.
- ▶ Smoothness can be measured by the size of  $|g''(x)|$ . Large values means the the function is rough (non-smooth).
- ▶ In order to obtain a smooth function we minimize

$$\sum_{i=1}^n (y_i - g(x_i))^2 + \underbrace{\lambda \int g''(t)^2 dt}_{\text{Roughness penalty}}$$

- ▶ It can be shown that the solution to this problem is a cubic spline with knots at  $x_1, x_2, \dots, x_n$ .
- ▶  $\lambda$  determined by cross-validation

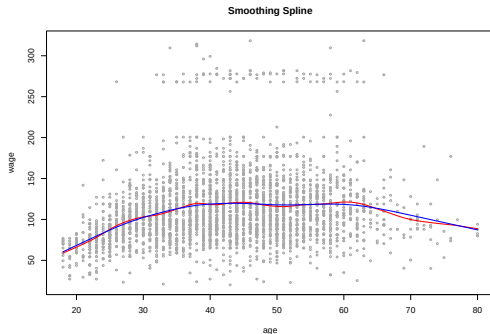
# Smoothing splines



# Example with Wage data

```
fit2=smooth.spline(age ,wage ,cv=TRUE)  
fit2$df
```

```
## [1] 6.794596
```



## Recap of kernel estimators



$$y = f(x) + \varepsilon$$

- ▶ Nadaraya-Watson estimator

$$\hat{f}(x_0) = \sum_{i=1}^n w_i y_i$$

where  $\sum_{i=1}^n w_i = 1$  and  $w_i = K(x_i - x_0)$

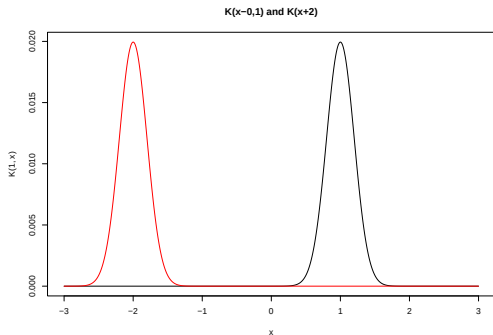
- ▶  $K$  is a so called kernel.
- ▶ Example of a kernel

$$K(x_i - x_0) = \frac{\phi\left(\frac{x_i - x_0}{h}\right)}{\sum_{i=1}^n \phi\left(\frac{x_i - x_0}{h}\right)}$$

where  $\phi$  is the standard normal density.

## Example of a kernel

```
K=function(x0,x,h=0.2) dnorm((x-x0)/h)/sum(dnorm((x-x0)/h))
```



If we want to predict  $y$  based for an observation with  $x = -2$  we should give high weights for observations with  $x$  close to  $-2$ .

## Generalized additive models (GAM)

- ▶ How do we extend the previously discussed methods to many predictors? Multiple non-linear regression? -

$$y = f(x_1, x_2, \dots, x_p, \varepsilon)$$

-or

$$y = f(x_1, x_2, \dots, x_p) + \varepsilon$$

- ▶ Usually not feasible to just extend the methods just described with these very general formulations.
- ▶ A simplification, the Generalized Additive Model (GAM)

$$y = \beta_0 + f_1(x_1) + f_2(x_2) + \dots + f_p(x_p) + \varepsilon$$



# Backfitting

- ▶ A method to use any of the models with one predictor and fit the GAMS model with many predictors.



Example with two predictors and the Nadaraya-Watson estimator

$$y = f_1(x_1) + f_2(x_2) + \varepsilon$$

- ▶ Sketch of procedure:
  1. Use one of the methods previously discussed to estimate  $f_1$  assuming  $f_2$  known, i.e. equal to zero.
  2. Estimate  $f_2$  based on the previous estimate of  $f_1$
  3. Estimate  $f_1$  based on the previous estimate of  $f_2$
  4. Iterate 2. and 3. until convergence.

## Backfitting

1. Start by assuming  $f_2(x_2) = 0$  and use the Nadaraya-Watson on

$$y = f_1(x_1) + \varepsilon$$

$$\implies \hat{f}_1(x_1)$$

2. Use Nadaraya-Watson on

$$\underbrace{y - \hat{f}_1(x_1)}_Z = f_2(x_2) + \varepsilon$$

$$\implies \hat{f}_2(x_2)$$

3. Use Nadaraya-Watson on

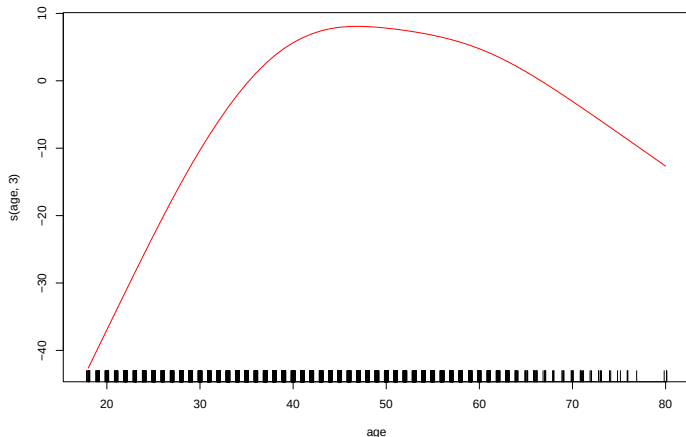
$$y - \hat{f}_2(x_2) = f_1(x_1) + \varepsilon$$

$$\implies \hat{f}_1(x_1)$$

4. Repeat 2-3 until convergence (until  $\hat{f}_1(x_1)$  and  $\hat{f}_2(x_2)$  does not change)

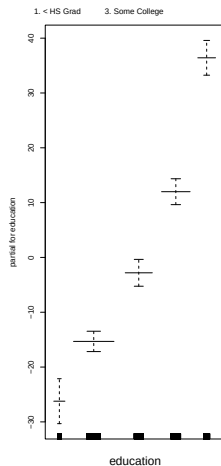
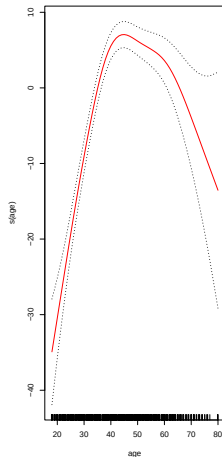
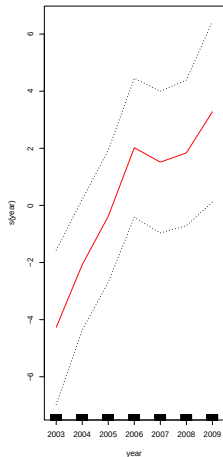
## Example with Wage data, GAM with smoothing splines

```
library(gam)
gam=gam(wage~s(year,3)+s(age,3),data=Wage)
plot.Gam(gam,terms="s(age, 3)",col="red")
```



# Possible to include linear terms

```
library(gam)
gam=gam(wage~s(year)+s(age)+education,data=Wage)
par(mfrow =c(1,3))
plot(gam,se=TRUE,col="red")
```



## GAM for classification (two-class case)

- ▶ We have previously discussed logistic regression where  $Y$ , conditionally on  $(X_1, \dots, X_p) = (x_1, \dots, x_p)$  is one with probability

$$p(\mathbf{x}) = P(Y = 1 | \mathbf{X} = \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}}$$

and zero otherwise.

- ▶ This implies that

$$\log \left( \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} \right) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

- ▶ Just as GAM gave us an extension to linear regression, it can do the same to logistic regression.
- ▶ ▶ This implies that

$$\log \left( \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} \right) = \beta_0 + f_1(x_1) + \dots + f_p(x_p)$$

## GAM for classification (two-class case)

```
gam_log=gam(I(wage>250)~s(year)+s(age)
            +education,data=Wage,
            family=binomial())
par(mfrow =c(1,3))
plot(gam_log,se=TRUE,col ="red")
```

